

# Data Science Toolbox Portfolio Questions

## 06 Perceptrons and Neural Networks

Daniel Lawson — University of Bristol

### Block 6

## Portfolio 06

### Portfolio Question 6.1.P - Gradient Descent

- We know that gradient descent let's us work out what direction to move in to decrease loss and improve performance. Explain backpropagation and its role in updating parameters through deep MLPs.
  - What approaches are used to improve training in practice?

### Portfolio Question 6.2.P - Attention

- Explain how a transformer works (in particular, what is attention?) and compare/contrast with how a CNN and a Vision Transformer process an image.

### Portfolio Question 6.3.P - Depth vs width?

- Investigate the limits of the Universal Approximation Theorem for MLPs, which says that an MLP with a single hidden layer of arbitrary width can approximate any continuous function.
- Take `model_1`, the first MLP from lab code and modify its architecture. First train it with 1 hidden layer with 64 neurons, compare this with 2 hidden layers with 32 each etc. for a variety of different configuration of hidden layers (make sure you have activation functions between, you might want to use a loop inside your class to make this easier to define).
- For a fixed number of epochs, record the training speed, model performance, minimum test loss etc. Produce some plots and explain your observations. You might want to watch this video for some inspiration: <https://youtu.be/qx7hirqgfuU?si=45VIYxsbazvJYv3b>
- Given a fixed budget of neurons, is it better to have an MLP that is wider or deeper?